

Energy Extraction from a Magic Formation

Background

You are an explorer of ancient ruins. Deep within the ruins, you discover a linearly arranged magical formation.

The formation consists of n energy nodes connected in a row. The i -th node contains an energy value of a_i .

There exists a mysterious energy interaction among these nodes. You may **activate and remove** the nodes one by one in any order to extract energy from the formation.

Energy Extraction Rules

When you activate the node with index i :

- The node immediately disappears from the formation;
- The energy you obtain is:

energy of the nearest existing node on the left $\times a_i \times$ energy of the nearest existing node on the right

If there is no node on one side, the energy on that side is considered to be 1.

After a node is removed, the formation shrinks, and the previously separated left and right nodes become adjacent.

Task

Design an activation order of the nodes such that the **total energy extracted** from the formation is maximized.

Output this maximum total energy.

Input Format

- The first line contains an integer n .
- The second line contains n integers $a_1, a_2, \dots, a_n$.

Output Format

Output a single integer representing the maximum total energy that can be obtained.

Sample Data

Sample Input 1

```
4
3 1 5 8
```

Sample Output 1

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167
```

Sample Explanation

One optimal activation order is: first activate the node with energy 1, then 5, then 3, and finally 8.

The energy gain process is as follows:

1. Activate 1: neighbors are 3 and 5, gain $3 \times 1 \times 5 = 15$
2. Activate 5: neighbors are 3 and 8, gain $3 \times 5 \times 8 = 120$
3. Activate 3: neighbors are 1 (boundary) and 8, gain $1 \times 3 \times 8 = 24$
4. Activate 8: no neighbors, gain $1 \times 8 \times 1 = 8$

Total energy:

$$15 + 120 + 24 + 8 = 167$$

Data Range

- $1 \leq n \leq 300$
- $1 \leq a_i \leq 10000$